Digital Forensics Case Handling, Autopsy and Sleuth Kit Analysis

OJE DOMINION MAYOWA

FWSD2511507

Computer and Digital Forensics

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29August 2026

LAB OVERVIEW

This lab focused on the **forensic examination and recovery of digital evidence** from a disk image while preserving its integrity. We learned to calculate **MD5 and SHA-256 hashes** to authenticate evidence, create and analyze a forensic case using **Autopsy**, and examine disk structures using **The Sleuth Kit (TSK)**.

Key activities included using mmls to identify partition offsets, fsstat to examine file-system information, and fls to locate files and deleted files. We used istat to examine file metadata and timestamps, icat to recover specific files such as **Income.xls**, and blkcat to inspect individual data blocks. We also used tsk_recover to recover multiple files from the forensic image.

Finally, recovered files were verified using **MD5/SHA-256 hash comparisons**, ensuring that the recovered evidence remained accurate and unchanged. Overall, the lab demonstrated the process of **preserving, analyzing, recovering, and validating digital evidence in a forensic investigation**.

OBJECTIVES

- Preserve Evidence: Calculate and verify MD5 and SHA-256 hashes to maintain data integrity.
- Perform Forensic Analysis: Analyze the disk image using Autopsy and identify relevant/deleted files.
- Analyze Disk Structure: Use TSK tools to examine partitions, file systems, sectors, and directories.
- Examine Metadata: Analyze file metadata, timestamps, and data block allocation using istat.
- Recover Evidence: Extract deleted files and data using icat, blkcat, and tsk_recover.
- Validate Recovery: Compare hashes of recovered files to verify data authenticity.
- Prepare Forensic Report: Document procedures, findings, errors, and evidence in a reproducible report.

TOOLS

- Terminal/Terminator
- Autopsy
- img_stat
- mmls, ffsstat, fls
- istat, icat
- blkcat, blkls
- tsk_recover
- md5sum, sha256sum.
- Sleuth Kit

METHODOLOGY

The authorised training forensic image was downloaded firstly to begin the lab

Fig 1.1 Downloaded Forensic Image

Next, we create a dedicated working directory named FORENSIC_INVESTIGATION_LAB1_OJE_DOMINION

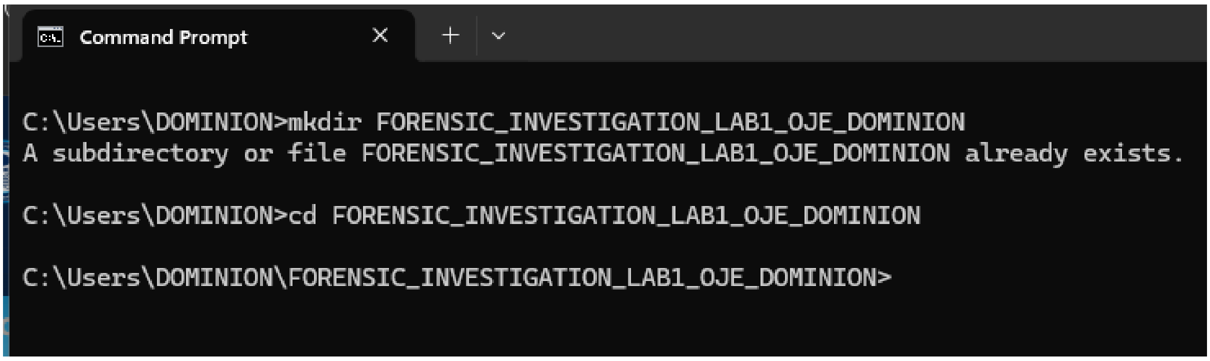


Fig 1.2 Lab Environment Created

PRE-ANALYSIS VERIFICATION TABLE

Evidence source	**Ch01InChap01.dd** (Authorized source image from the ICDFA portal)
File size	1440KB
Storage location	C:\Users\DOMINION\FORENSIC_INVESTIGATION_LAB1_OJE_DOMINION\Evidence
MD5 hash	a117773bcf1fc88ec0ab8e0a349fbbcb
SHA-256 hash	3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122

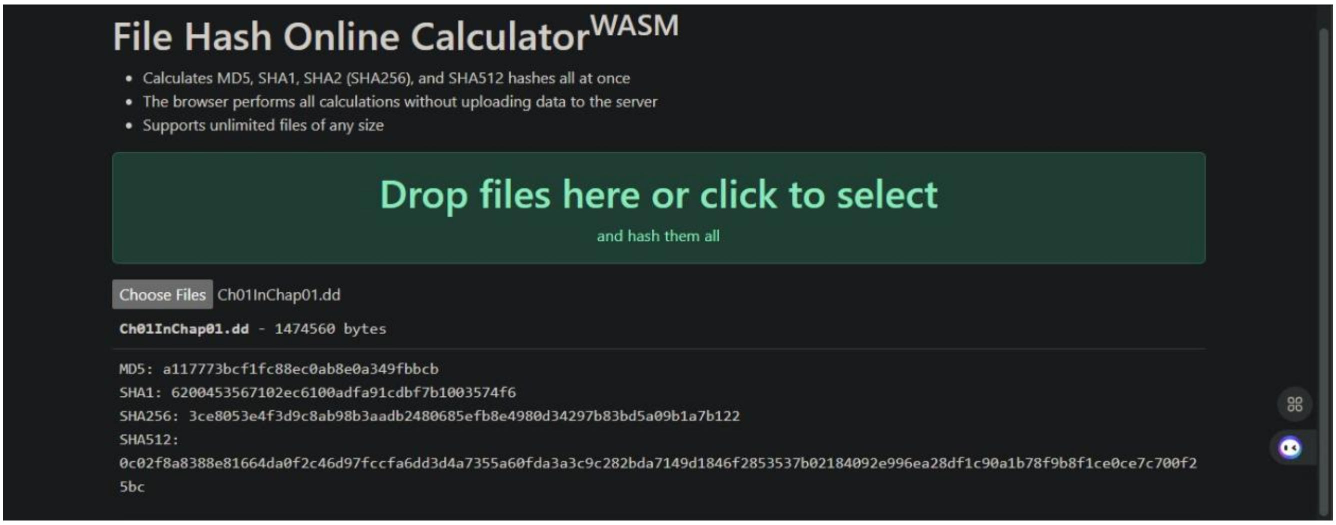


FIG 1: Online checking of the different Encryption of the source file.

PART A

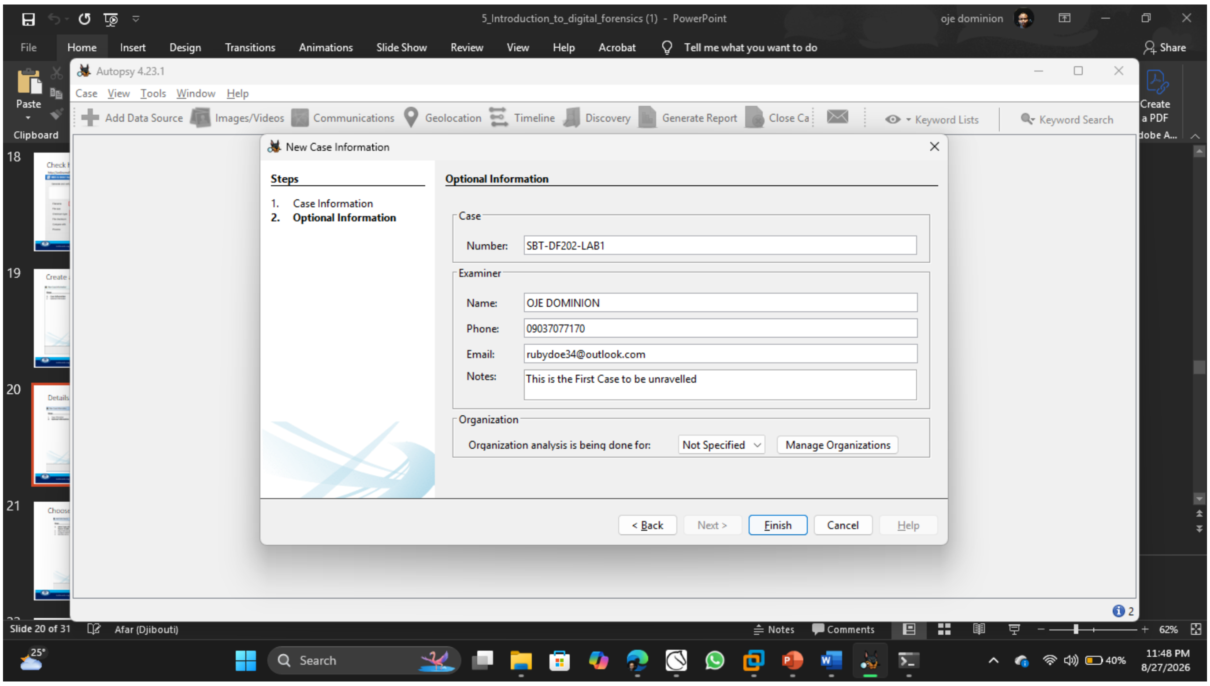


Fig 2: The First Case Created on Autopsy

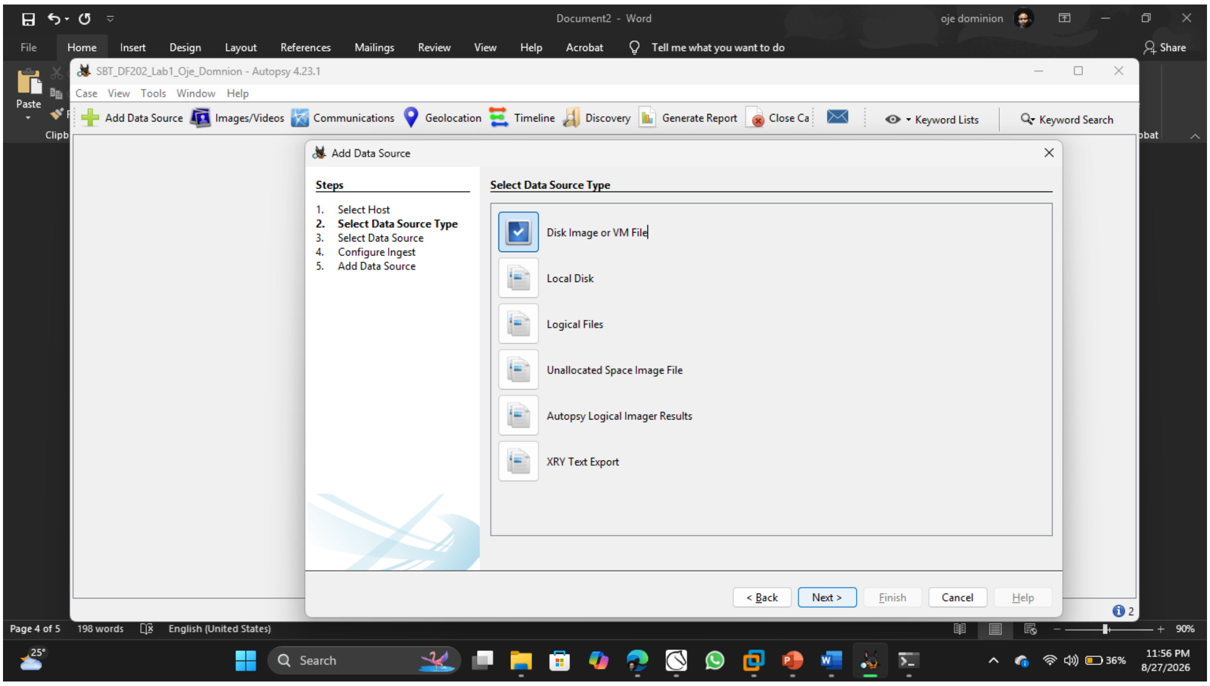


Fig 3: Setting up the Case Lab on Autopsy

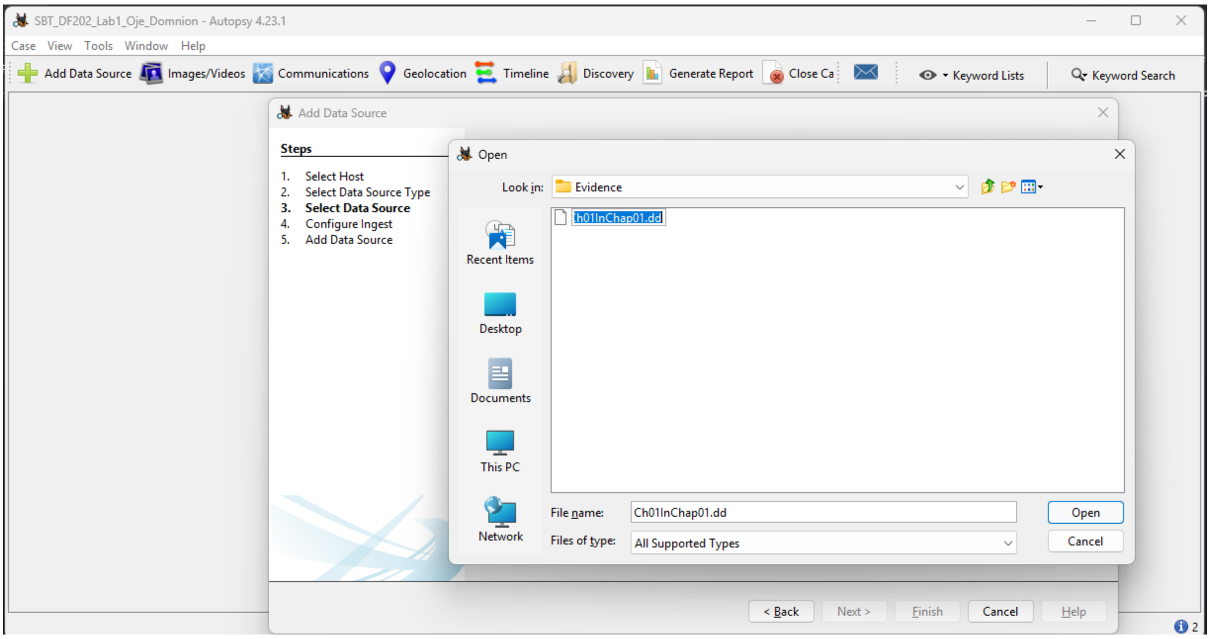


Fig 4: Selecting the Source Image or File

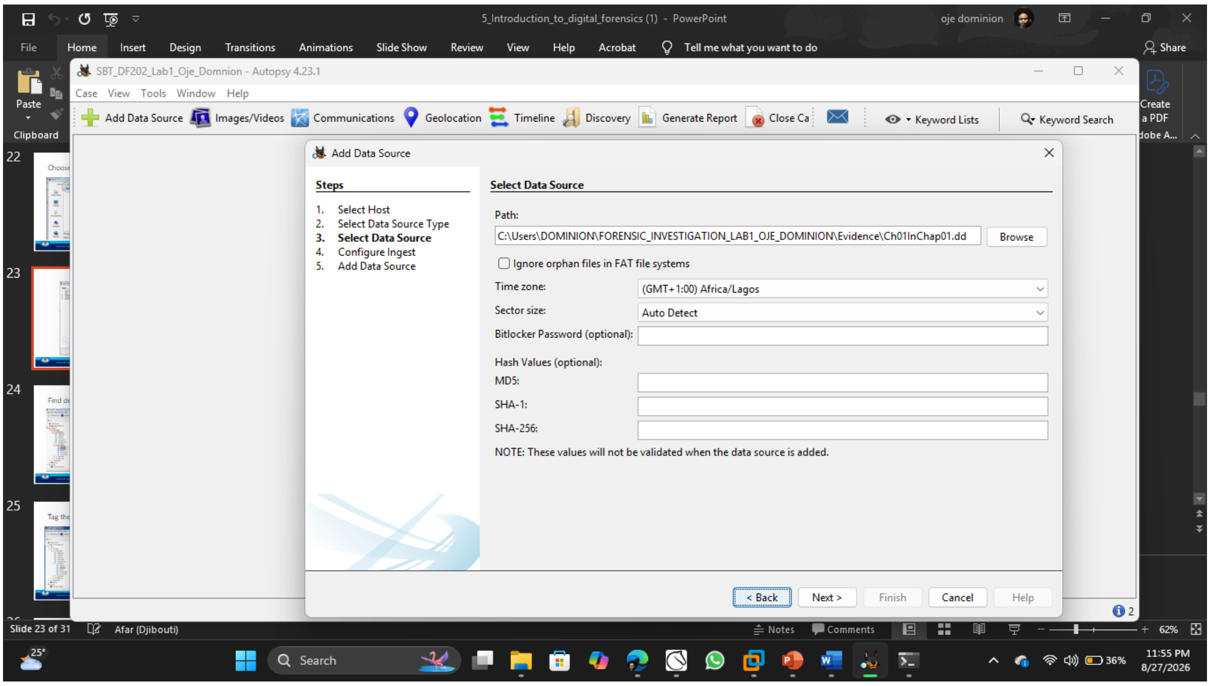


Fig 5: The First Case Created on Autopsy

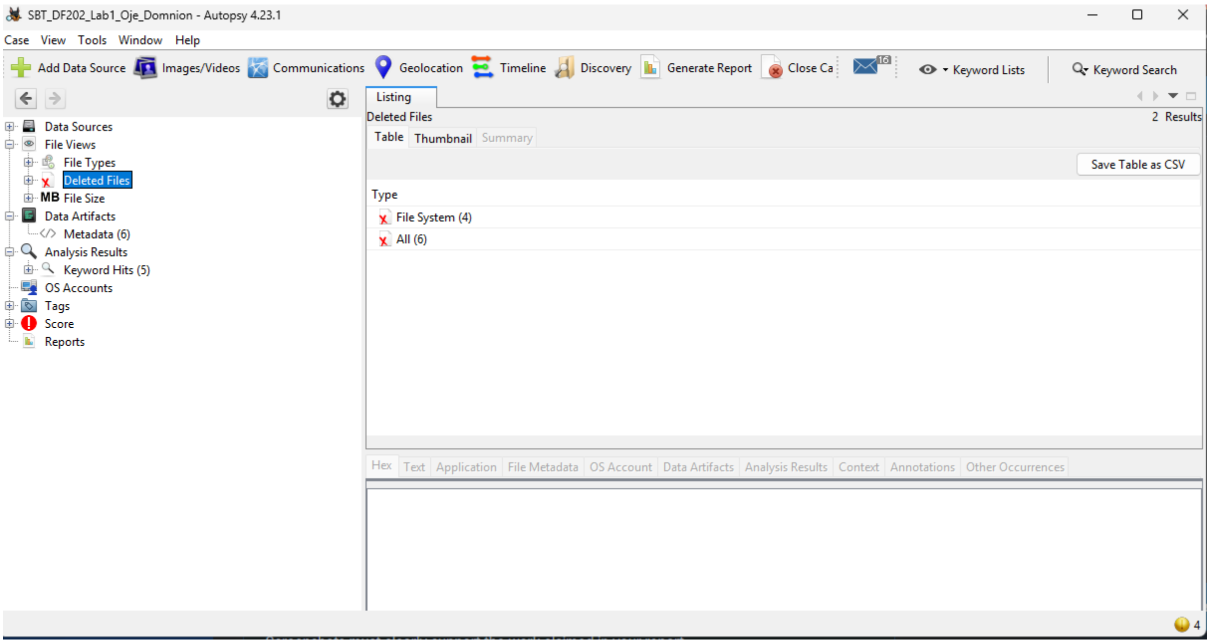


Fig 6: List of files in the system

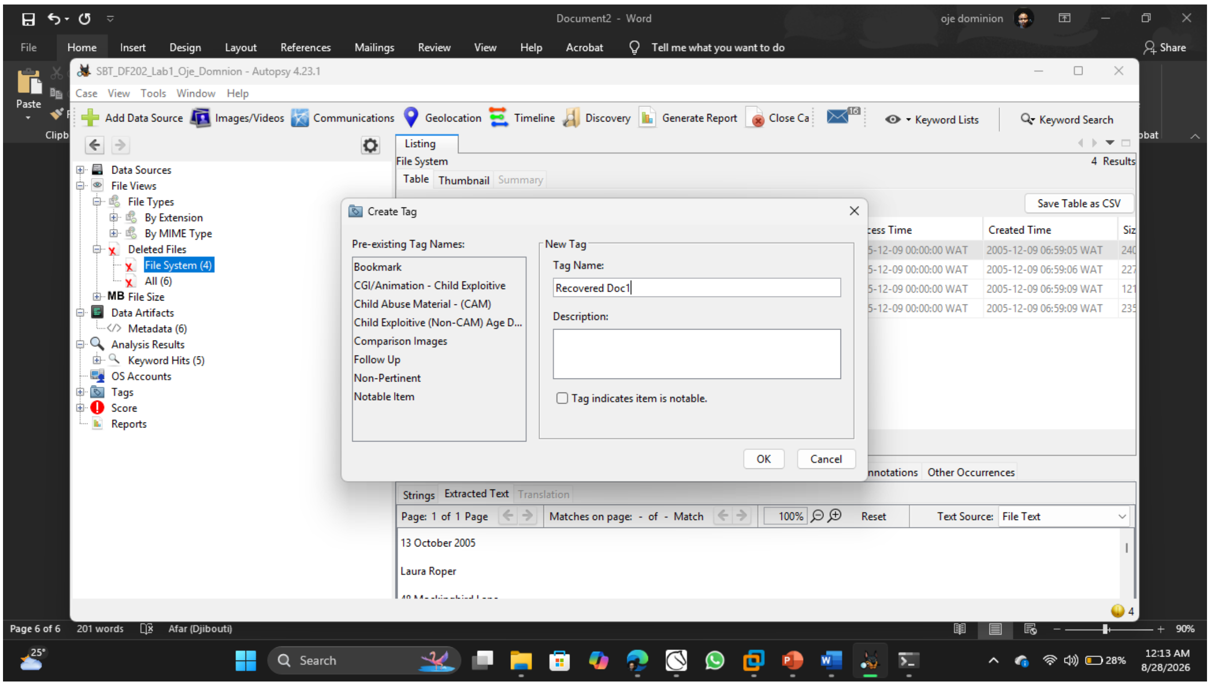


Fig 7: Tagged Recovered File

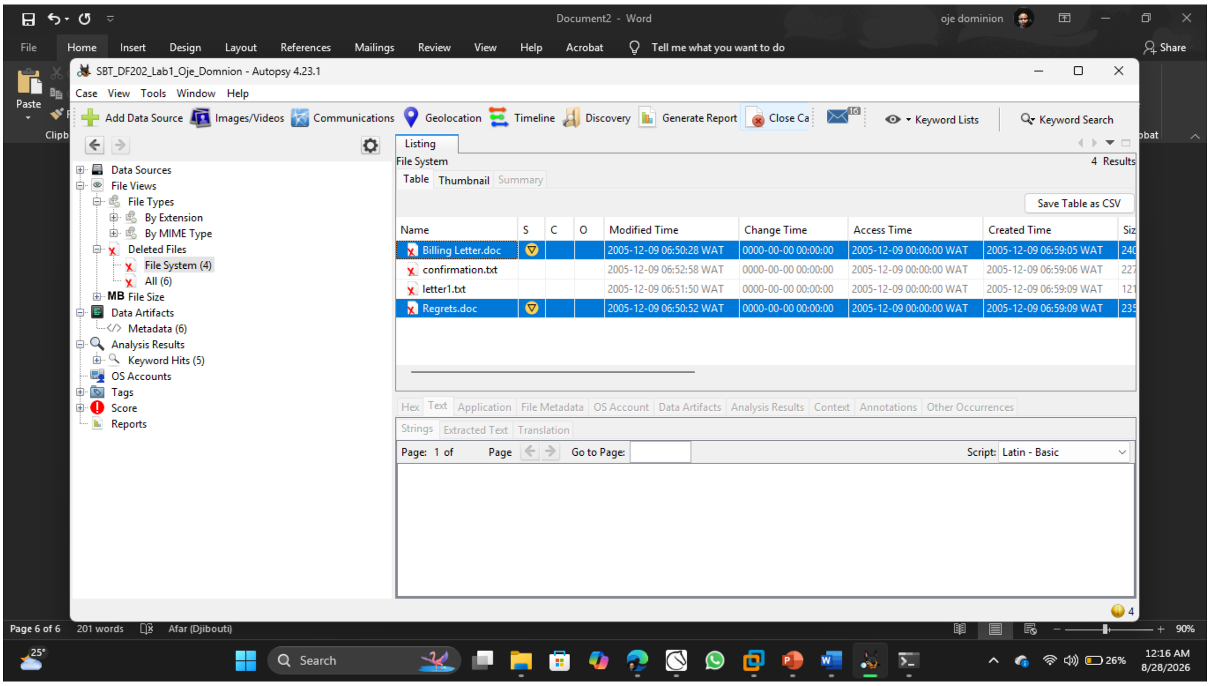


Fig 8: List of Tagged files in the system

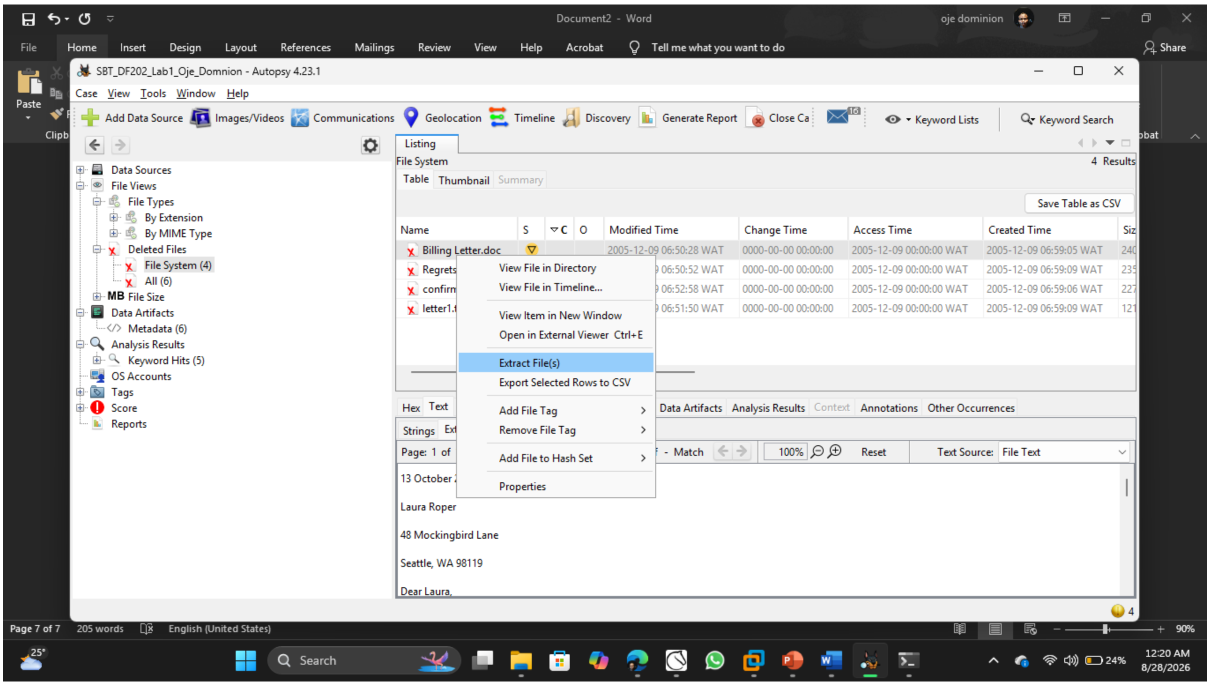


Fig 9: Extracting of files in the system for Recovery

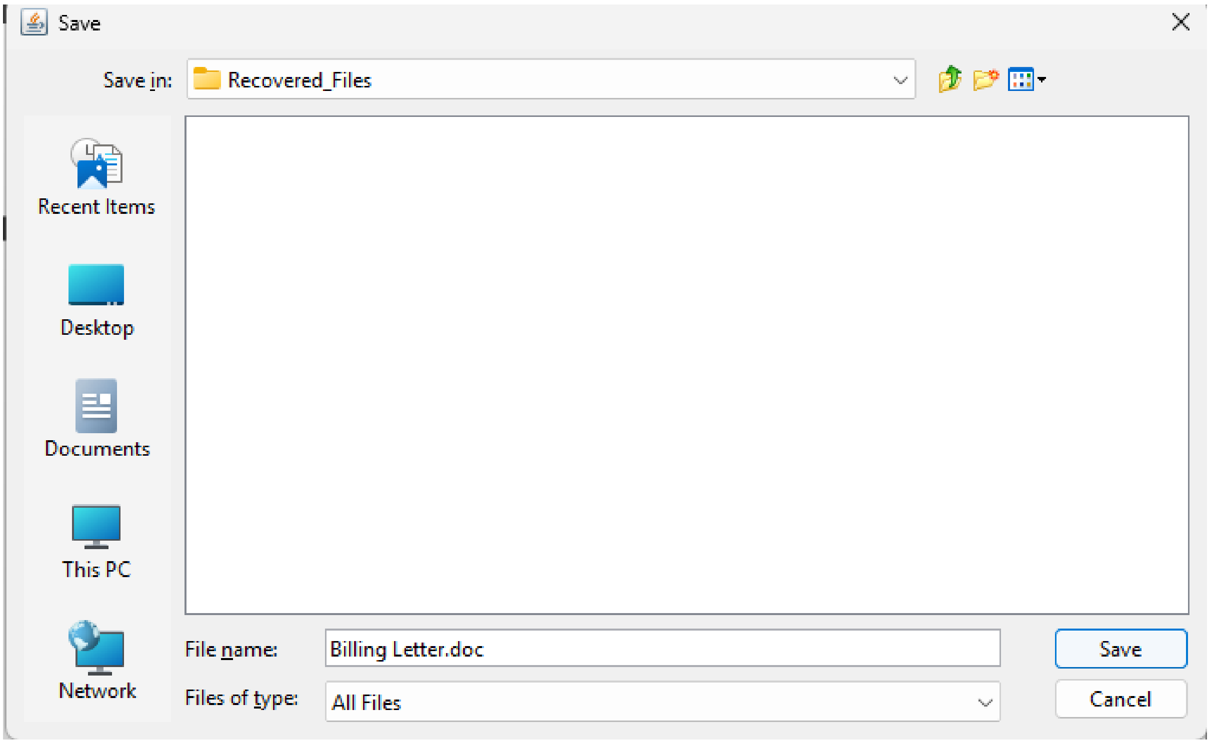


Fig 10: Recovered Files Pane

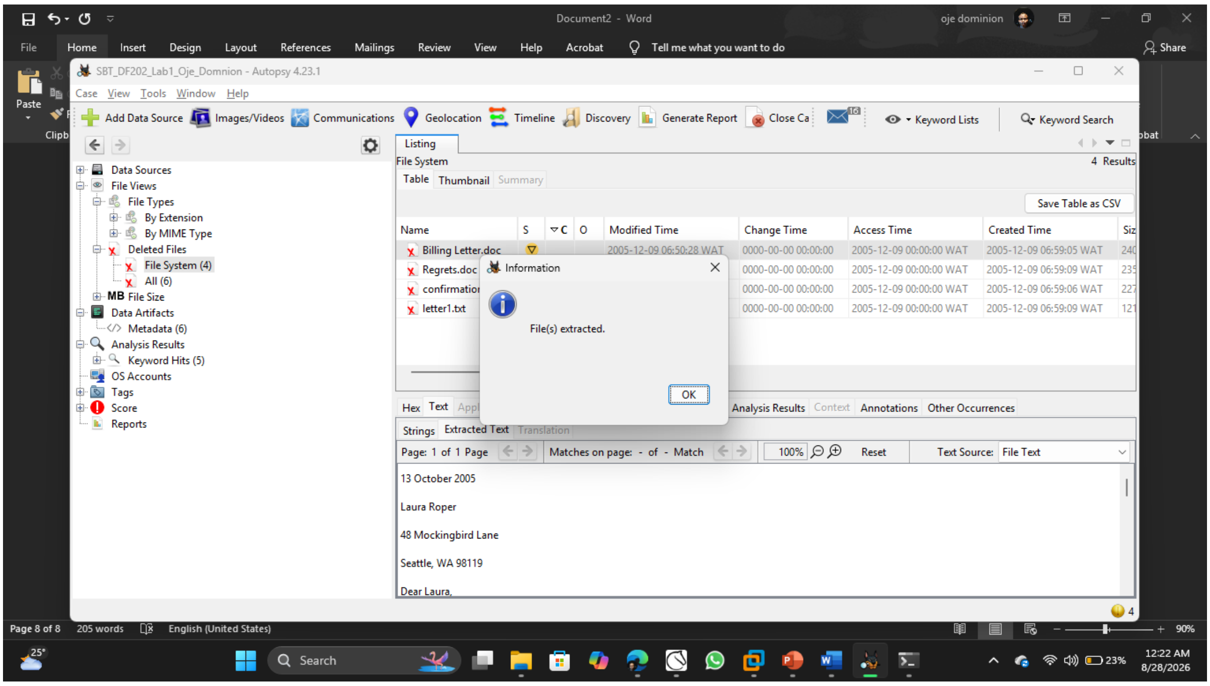


Fig 11: Files Recovered Successfully

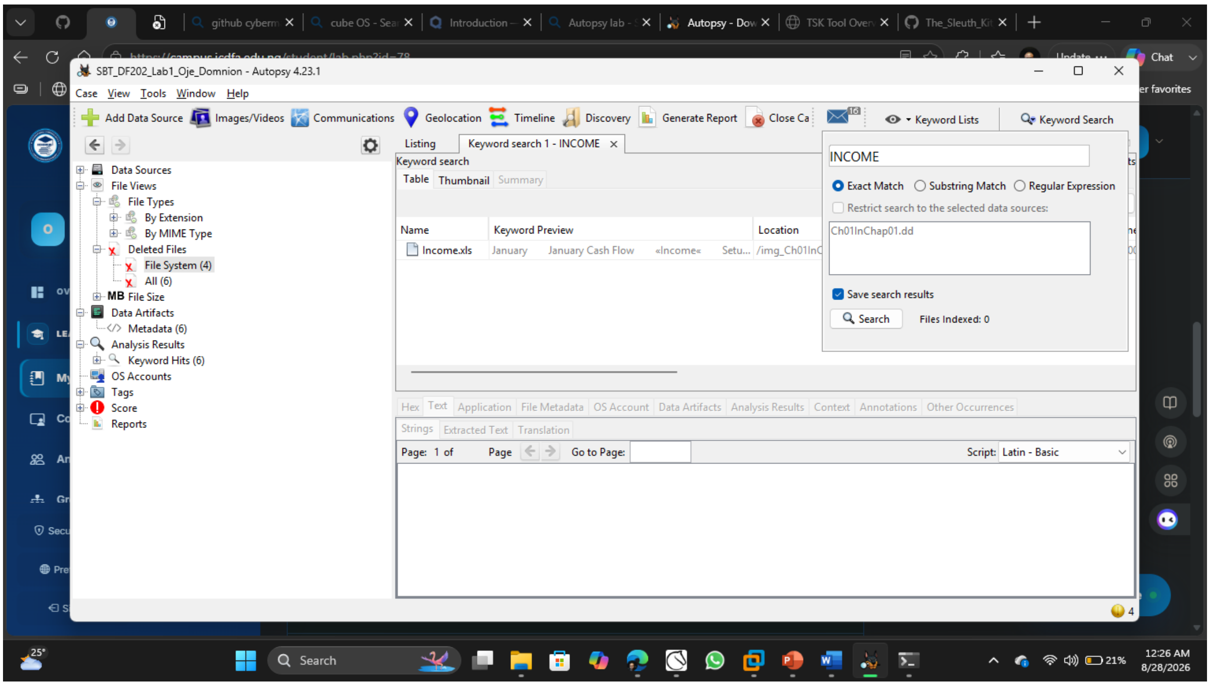


Fig 12: Keyword Search Performed on Autopsy

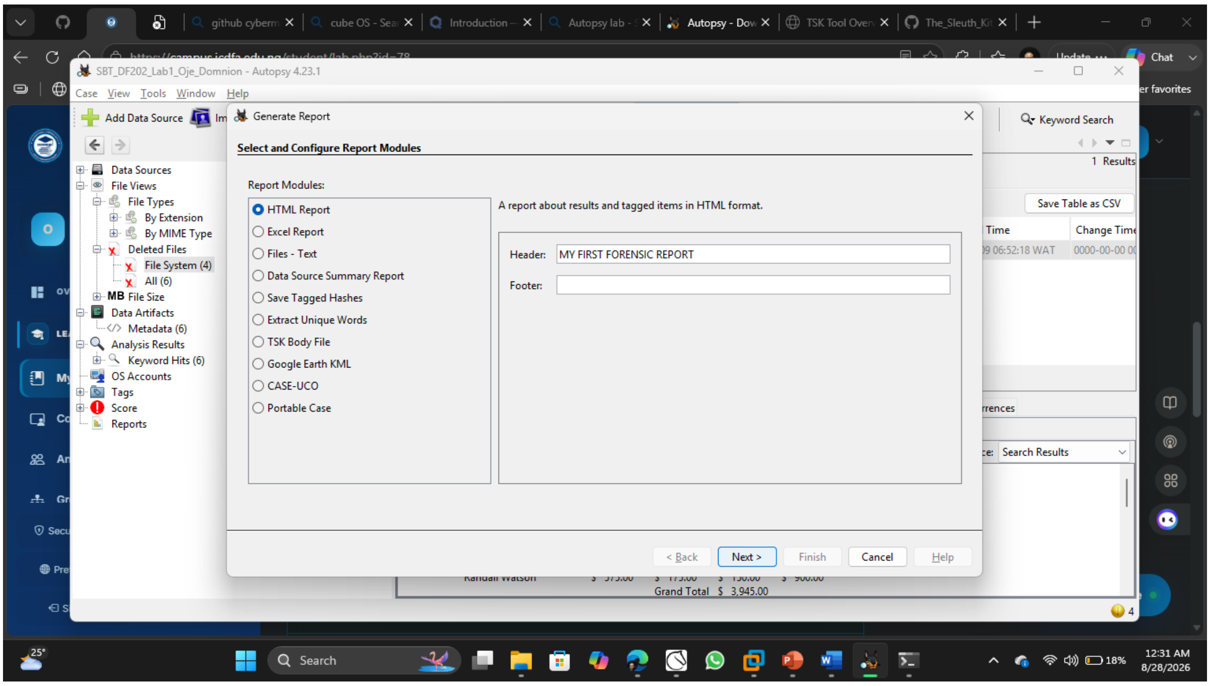


Fig 13: Generating Report from the case

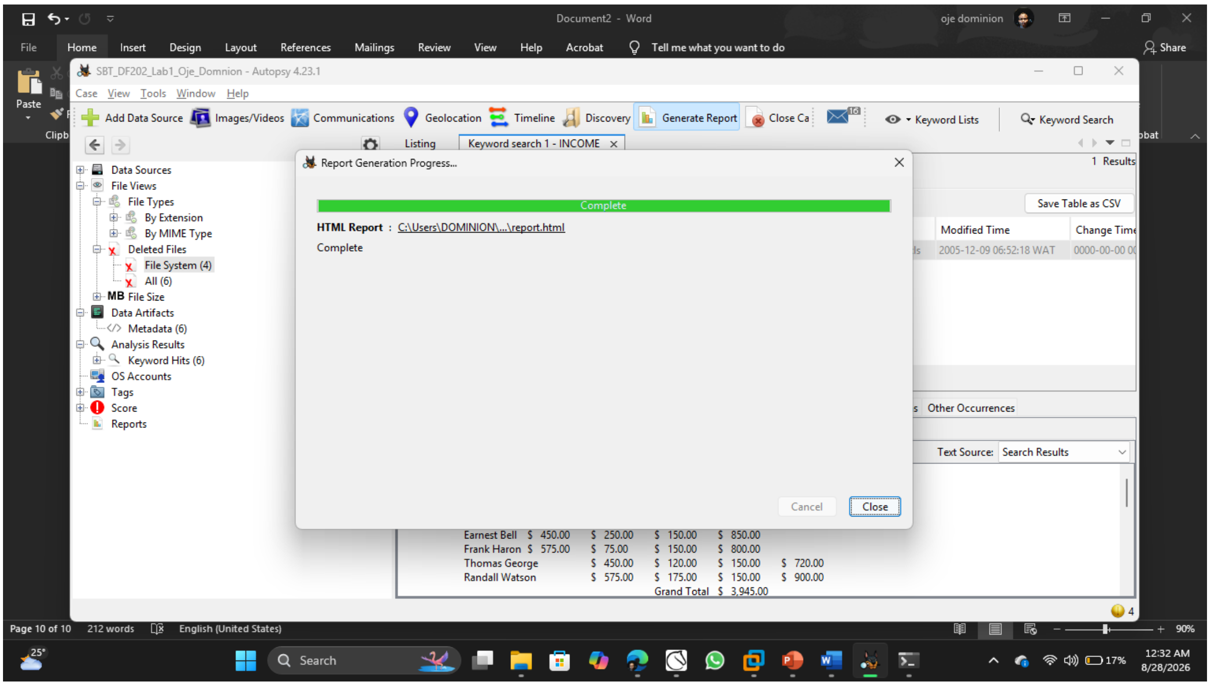


Fig 14: Reports Generated to be viewed as an HTML File

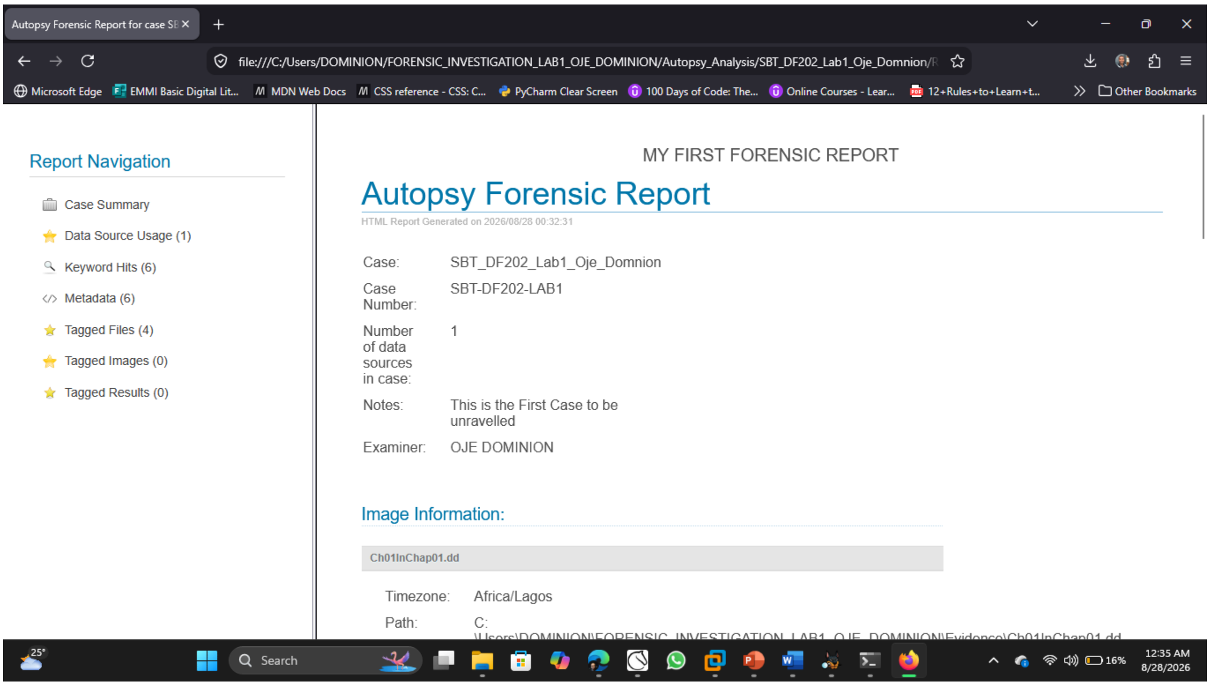


Fig 15: HTML view of the Generated Forensic Summary Report

FORENSIC ANALYSIS SUMMARY TABLE

case name	SBT_DF202_Lab1_Oje_Domnion
evidence source	C:\Users\DOMINION\FORENSIC_INVESTIGATION_LAB1_OJE_DOMINION\Evidence\Ch01InChap01.dd
deleted-file findings	/Ch01InChap01.dd/Billing Letter.doc /Ch01InChap01.dd/Regrets.doc / Ch01InChap01.dd/Confirmation.txt / Ch01InChap01.dd/letter.txt
recovered files	/img_Ch01InChap01.dd/Income.xls /img_Ch01InChap01.dd/Billing Letter.doc /img_Ch01InChap01.dd/Regrets.doc
keyword-search results	/img_Ch01InChap01.dd/Income.xls
reporting evidence	file:///C:/Users/DOMINION/FORENSIC_INVESTIGATION_LAB1_OJE_DOMINION/Autopsy_Analysis/SBT_DF202_Lab1_Oje_Domnion/Reports/SBT_DF202_Lab1_Oje_Domnion HTML Report 08-28-2026-00-32-31/report.html

Part B - Sleuth Kit Analysis

The first step is to create a usb directory where the evidence image would be stored and then to download the evidence image into the directory using wget.

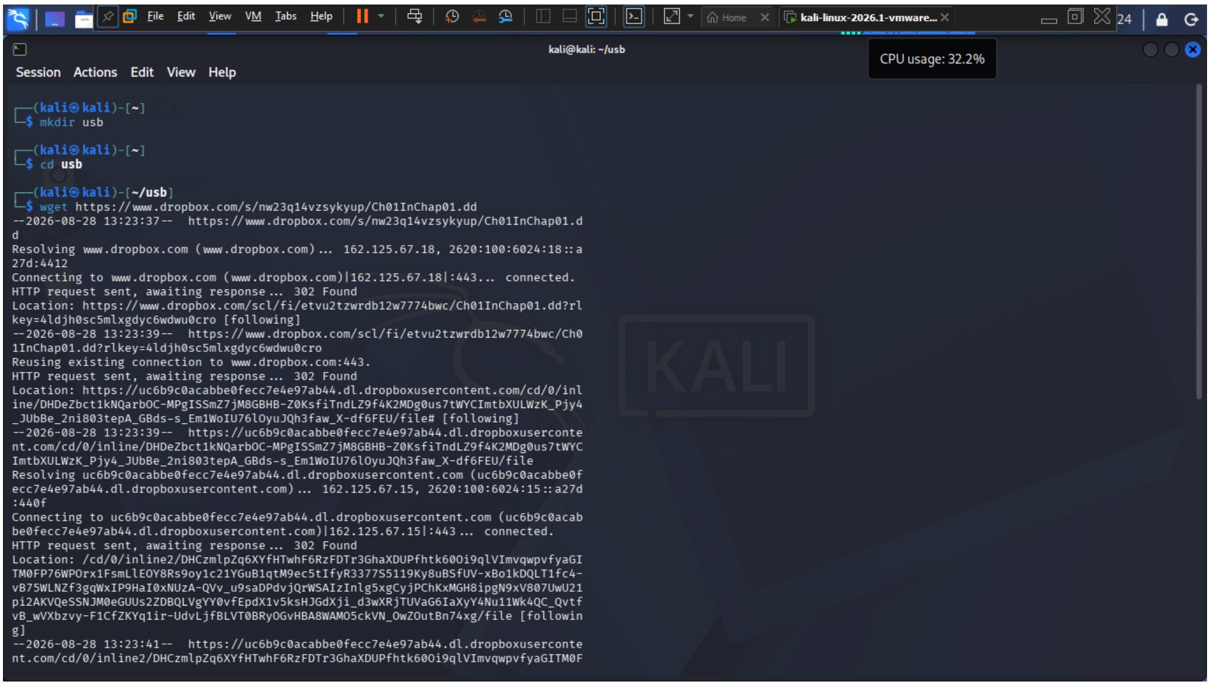


Fig 1.2 Setting up the USB Directory

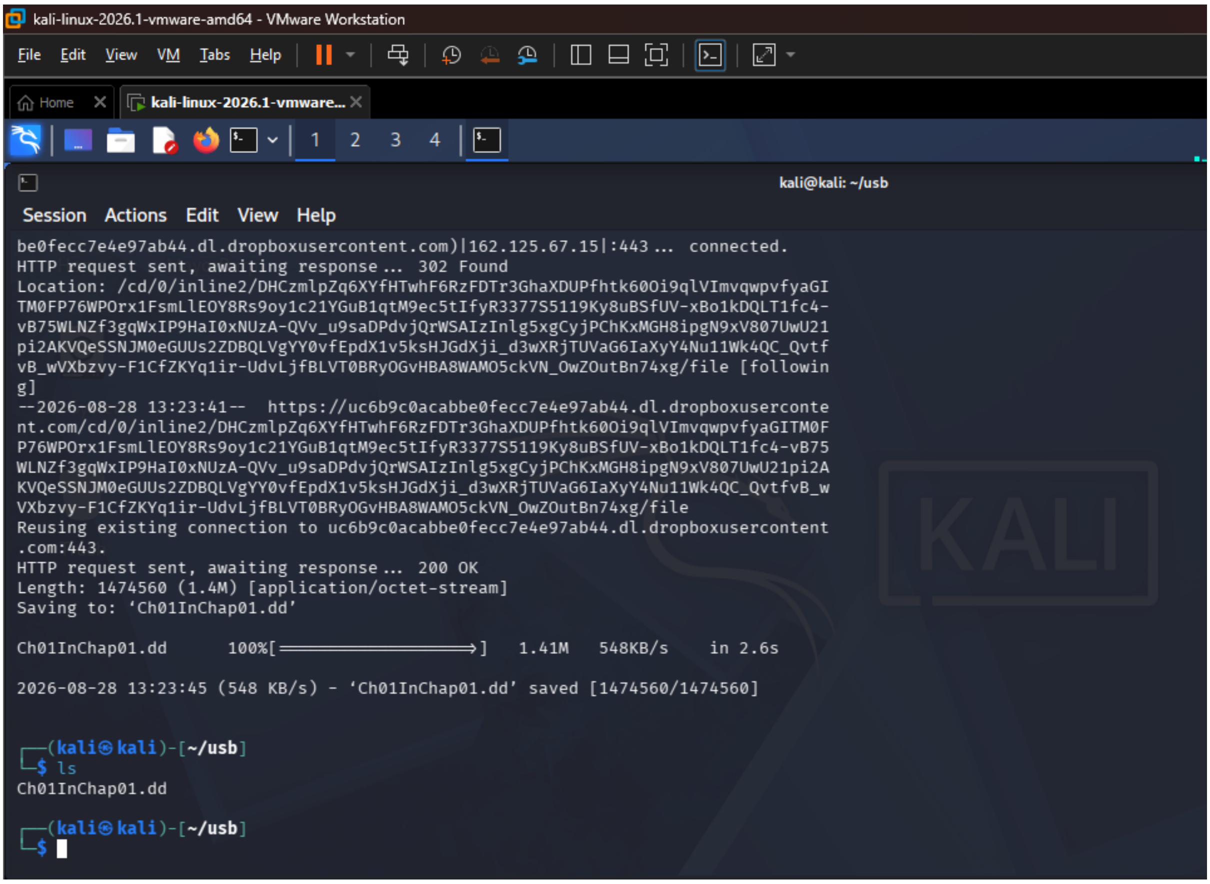


Fig 1.3 Disk Image confirmed in the USB Directory

Then we do a little statistical summary analysis of the image source using img_stat

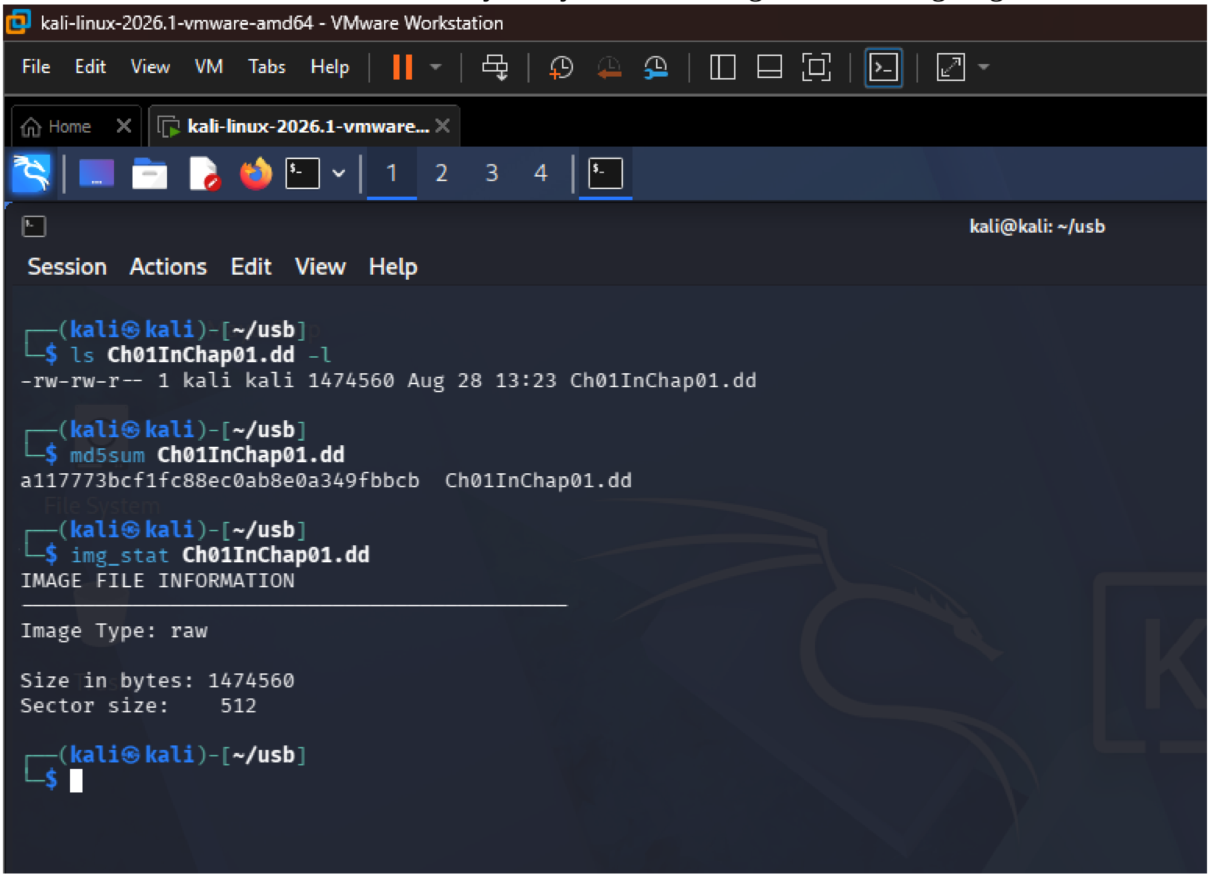


Fig 1.4 Summary analysis of the Forensic Image

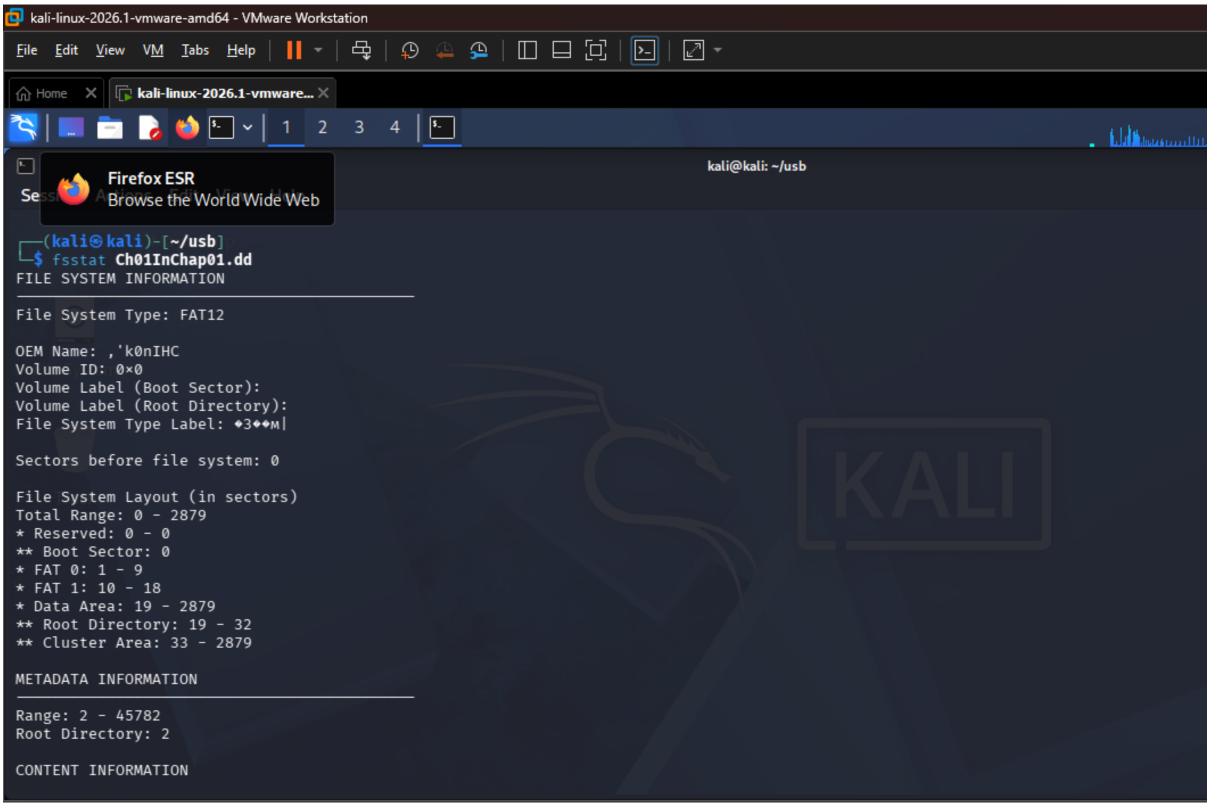


Fig 1.5 Showing Files system Details of the Source Image

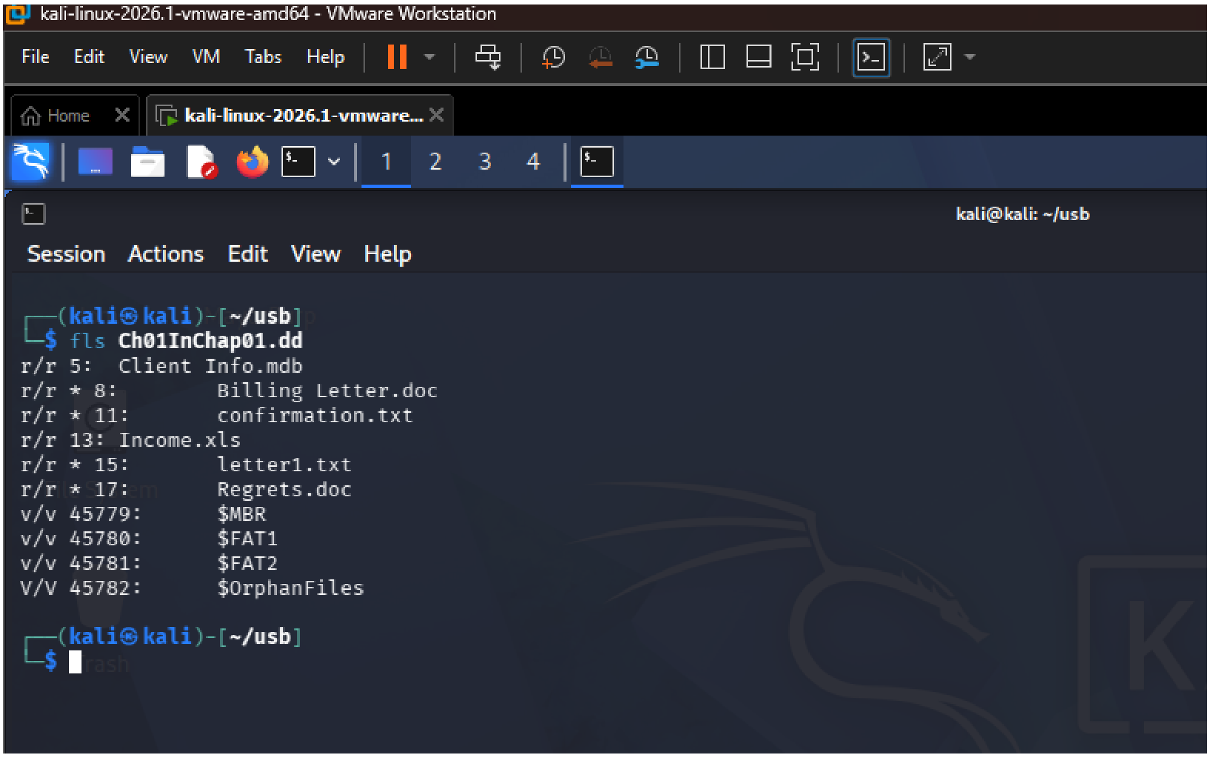


Fig 1.6 Showing the List of allocated and deleted File Names in the Directory

In Fig 1.6 above, here we see the INCOME.XLS File.

Fig 1.7 Displaying Deleted Entries only.

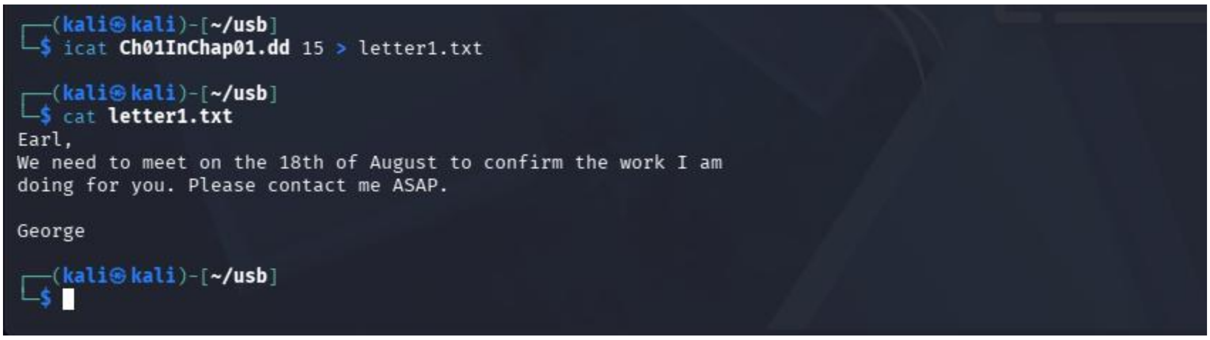


Fig 1.8 Displaying Deleted Entries only.

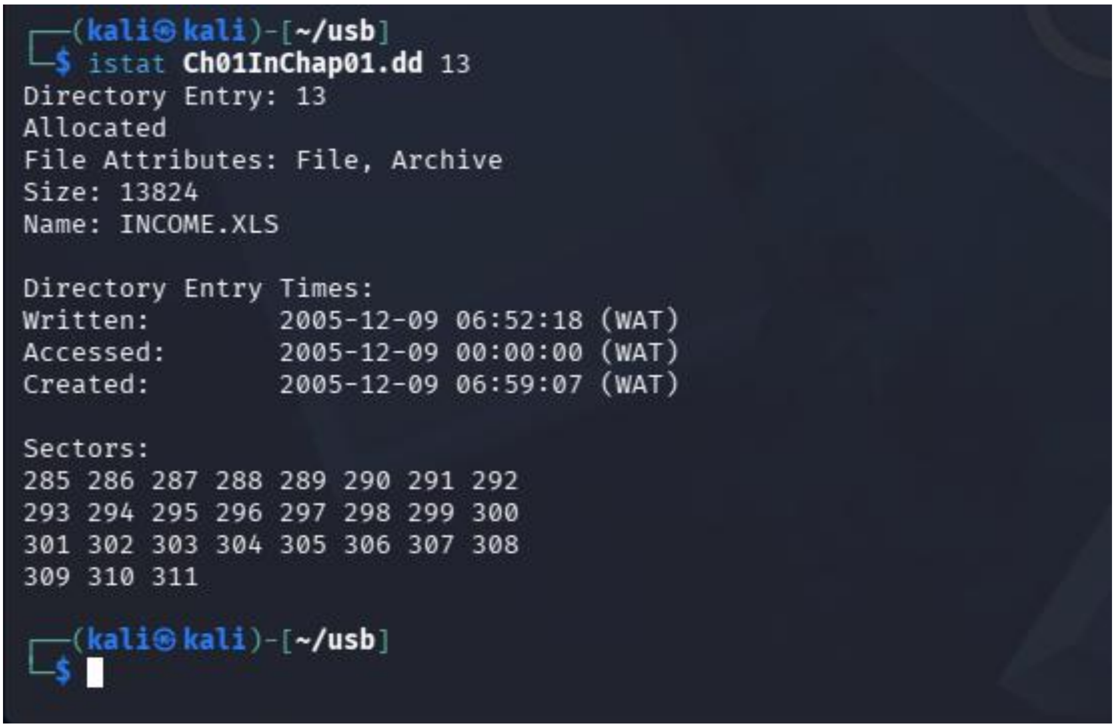


Fig 1.9 Displaying the statistics about the files Metadata

This shows that this is a file with multiple sectors, Next we extract from the file sectors number

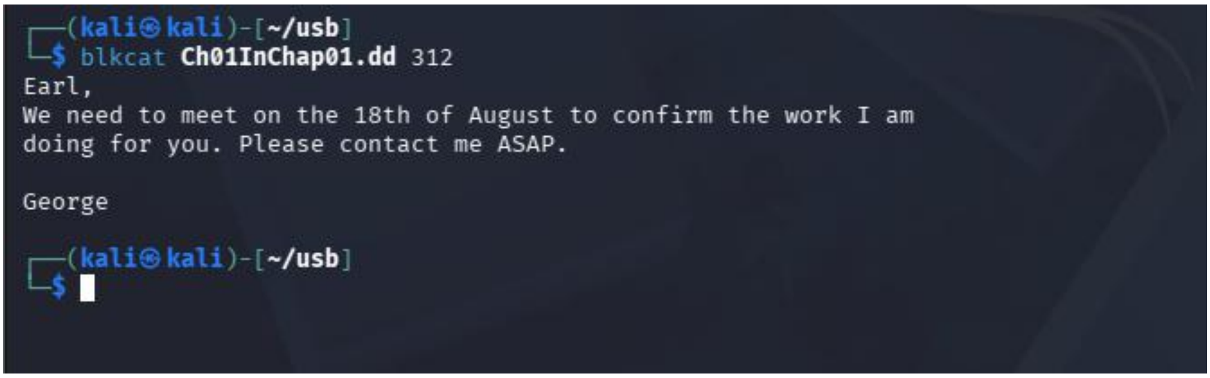


Fig 1.10 Extracting the Content of the file from the Sectors Number using blkcat

Some of the commands used and their uses are:

1. **Istat:** This examines metadata and timestamps of a specific file/inode.
2. **Fls:** Lists files and directories, including deleted files, in a disk image.
3. **Blkcat:** Displays the contents of a specific data block in the file system.
4. **Icat:** Recovers/extracts a specific file from a disk image using its inode.
5. **Fsstat:** Displays detailed information about the file system structure and metadata.

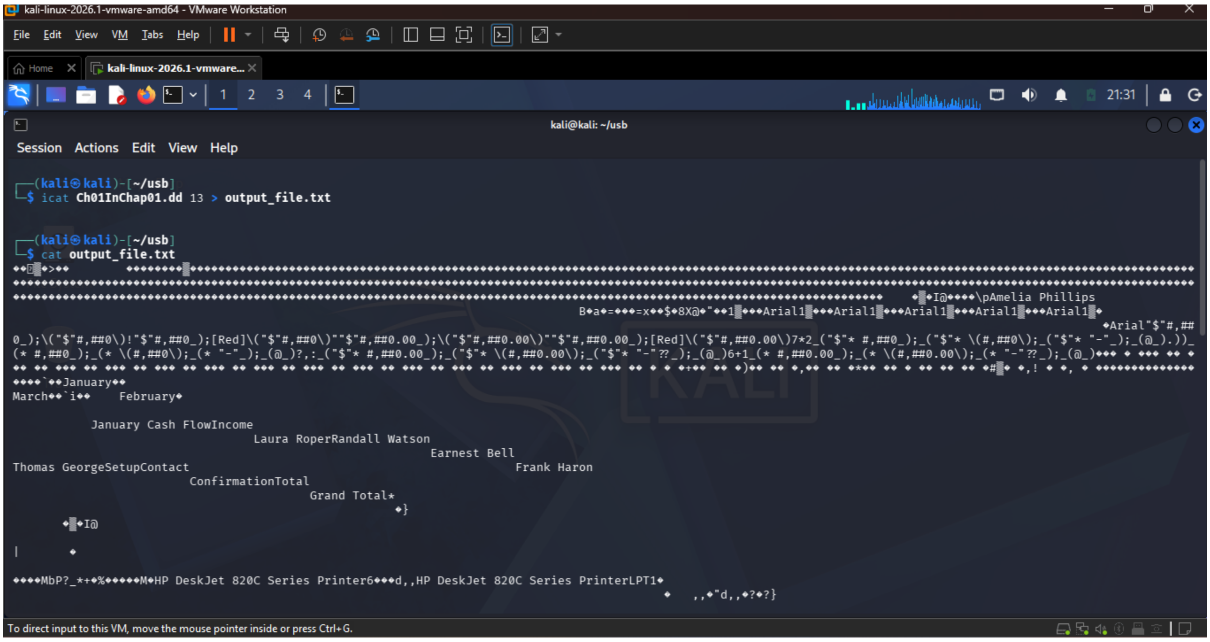


Fig 1.11 Recovering Income.xls using icat

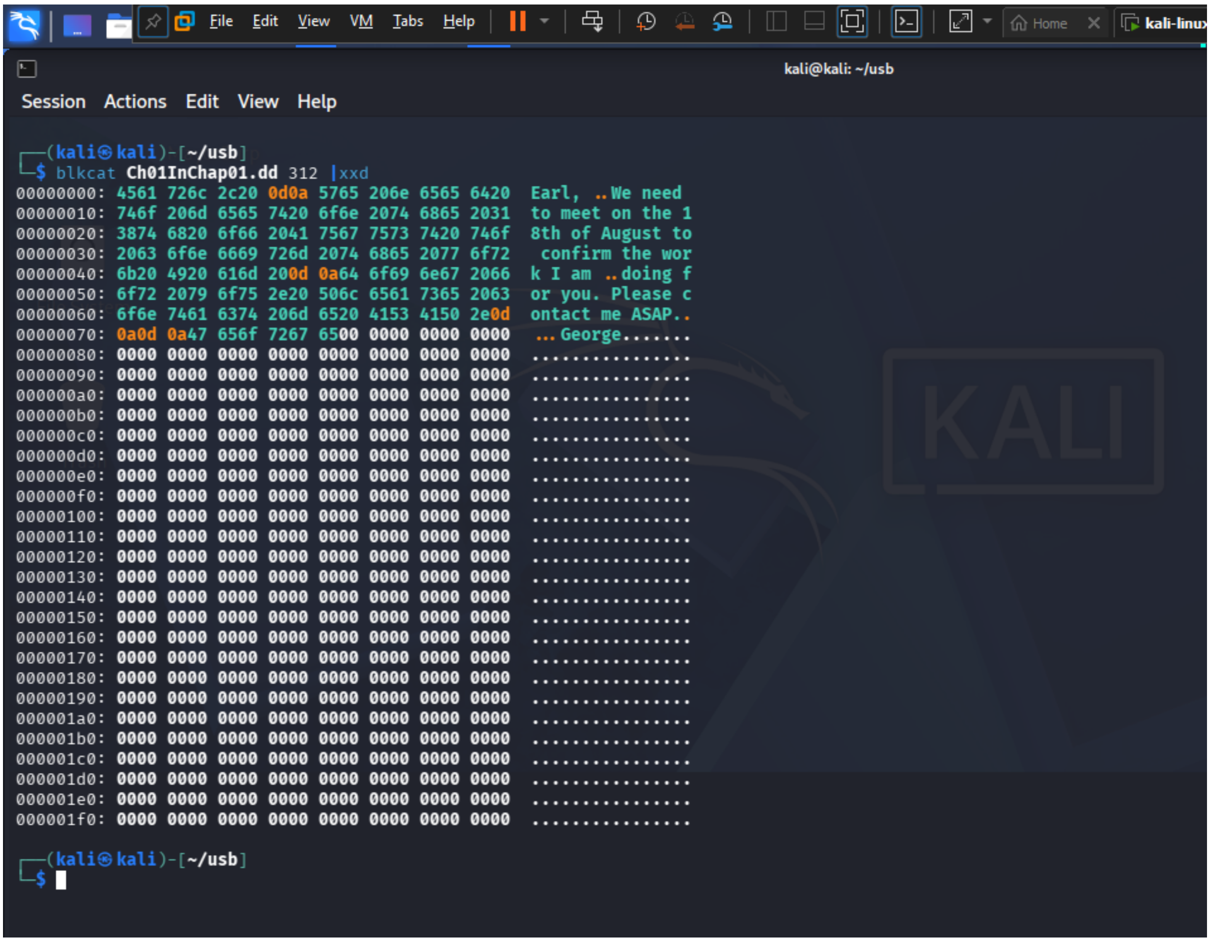
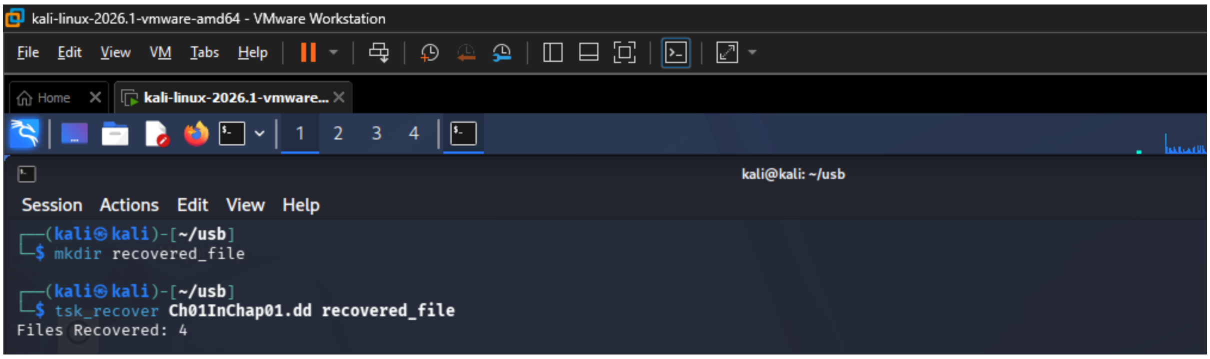


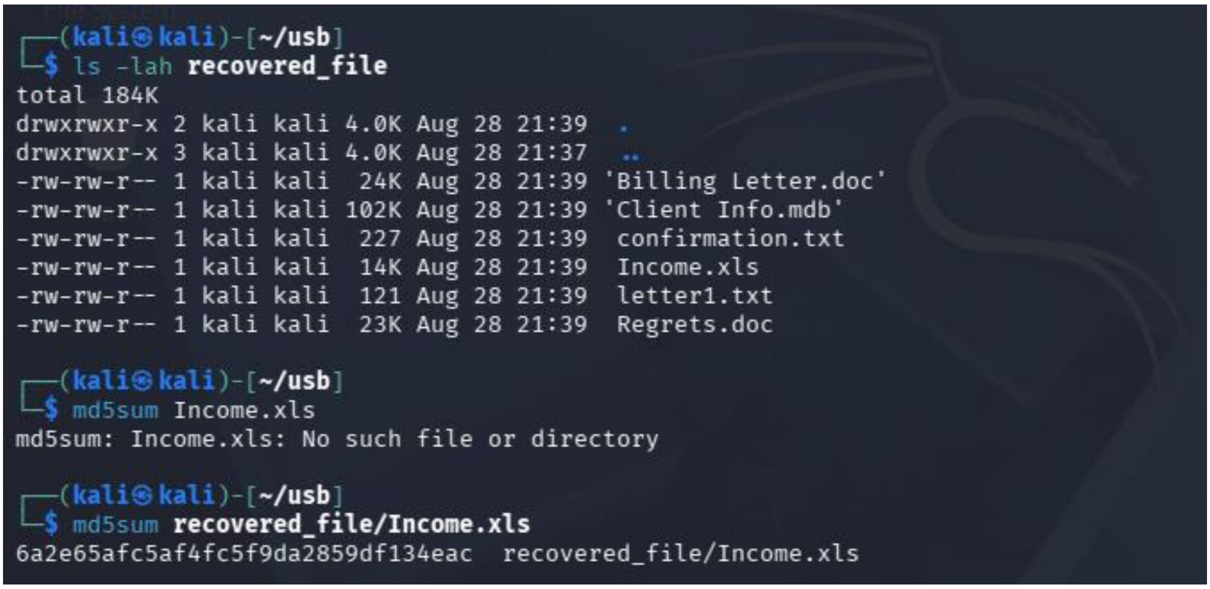
Fig 1.12 Extracting the content of the disk image in Hex

Using Tsk_Recover to recover deleted files, we first create a directory named recovered_file

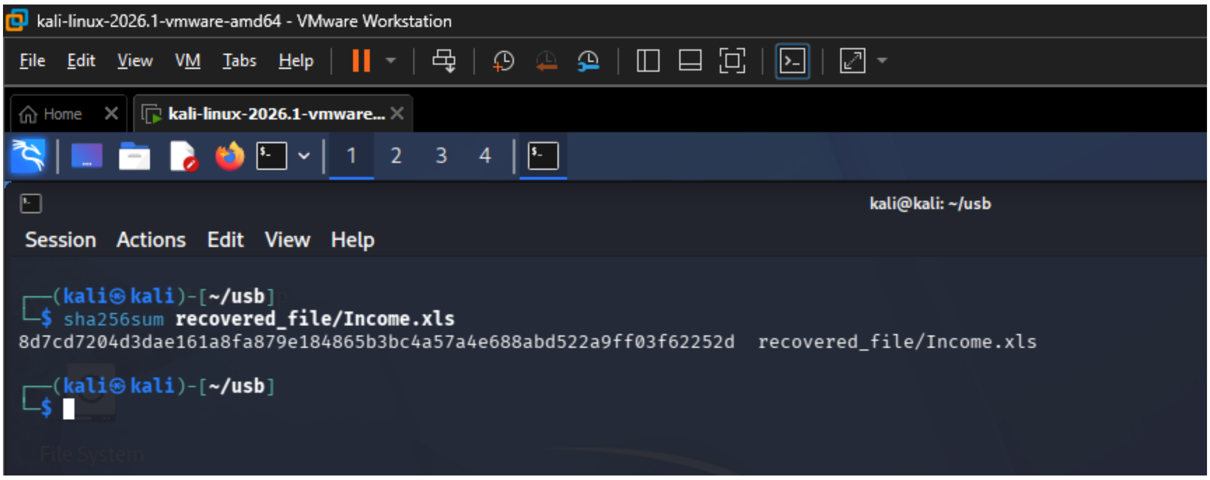


When used without any option we recovered 4 files in total, but when used with the option -e it recovers both the allocated and the deleted files thereby recovering a total of 6 files.

Now we compare the Hashes of the recovered File
For MD5Sum we have below,



And for the Sha256sum we have,



In conclusion, the lab provided practical experience in preserving, analyzing, recovering, and validating digital evidence using Autopsy and The Sleuth Kit. It demonstrated how forensic tools and hash verification can ensure the integrity and reliability of recovered evidence during a digital investigation.